
Tom Banker

Technical Artist & Developer

San Francisco, CA

tombanker@gmail.com

EXPERIENCE

April 2022 – Present
Cupertino, CA

Apple — R&D Technical Artist

- Developed and shipped System Environments [Jupiter, Thórsmörk] for visionOS, writing custom Metal shaders to raise realtime visual fidelity within GPU budget [11ms]
- Contributed graphics engineering and rendering work to shipping products and features [MacOS preview raytracer]
- Prototyped R&D rendering techniques and built internal SwiftUI/RealityKit apps and demos for the executive team

April 2020 – April 2022
Menlo Park, CA

Meta — Technical Artist

- Built real-time AR effects for Instagram and Messenger in SparkSL (GLSL), from LUTs and PBR shaders to complex particle effects, optimized for a wide range of mobile devices
- Researched, prototyped, and pitched new effects, including ML style-transfer, using the latest real-time techniques
- Gave optimization feedback to third-party vendors, wrote internal documentation, and onboarded and mentored junior technical artists

July 2017 – April 2020
Palo Alto, CA

Machine Zone (AppLovin) — Technical Artist

- Led development of the asset pipeline that transitioned MZ's games from 2D to 3D; supported and shipped titles including Crystalborne: Heroes of Fate and World War Rising
- Designed SQL databases for asset tracking and game binary size optimization
- Built and maintained LiveOps sales tools and wrote Jenkins CI scripts for engine updates and game builds

January 2016 – July 2017
San Francisco, CA

Tinyco (Jam City) — Technical Artist

- Led the 3D art pipeline and tools for the art department, finding and removing bottlenecks between artists and engineers
- Managed a repository of 4,000+ rigged assets for Marvel Avengers Academy and R&D rigging for Harry Potter: Hogwarts Mystery
- Assisted engineers with shader implementation and engine optimizations

April 2015 – October 2015
Irvine, CA

Blizzard Entertainment — Cinematics
Rigging and Simulation Technical Director (TD)

- Rigged characters and props for the WoW: Legion, Heroes of the Storm, and Overwatch cinematics
- Set up cloth and character simulations and post-simulation sculpting

June 2013 – April 2015
San Francisco, CA

Industrial Light and Magic (ILM) — Creature Technical Director (TD)

- Built rig and simulation setups (muscles, flesh, rigid body, cloth) for creatures and props on Jurassic World, Teenage Mutant Ninja Turtles, and Noah
- Contributed to Block Party 2 (Academy Technical Achievement Award) and wrote creature pipeline scripts and tools, including the asset validation system
- Supported artists across departments through shot and asset troubleshooting

April 2012 – April 2013
San Francisco, CA

LucasArts — Associate Character Technical Director (TD)

- Started development of the rigging, animation, and cinematics pipelines and tools for Star Wars 1313 to improve artist efficiency
- Rigged real-time props, characters, and creatures, and worked with ILM to convert offline cloth simulations to real-time
- Contributed to projects led by the Advanced Development Group (ADG)

June 2011 – August 2011
Culver City, CA

Sony Pictures Imageworks — Character Technical Director (TD) Intern

- Rigged creatures and props and built deformations for The Amazing Spider-Man
- Created training resources for the Rigging Department

SKILLS

Languages	C++, Python, Swift, Metal Shading Language, GLSL, HLSL
Graphics & GPU	Metal (render and compute), real-time rendering, PBR, path tracing, render pipeline architecture, shader authoring, GPU performance debugging & optimization
Frameworks & tools	RealityKit, ARKit, SwiftUI, MetalKit, Reality Composer Pro, Xcode GPU capture and Instruments
Content & pipeline	USD, MaterialX, OpenPBR, Houdini, Maya, Blender, Jenkins CI, SQLite

EDUCATION

September 2011
Savannah, GA

Savannah College of Art and Design (SCAD)
MFA Animation

May 2009
West Lafayette, IN

Purdue University
B.S. Computer Graphics Technology